

Neural and Evolutionary Computation (NEC)

A2: Optimization with Genetic Algorithms

Objective

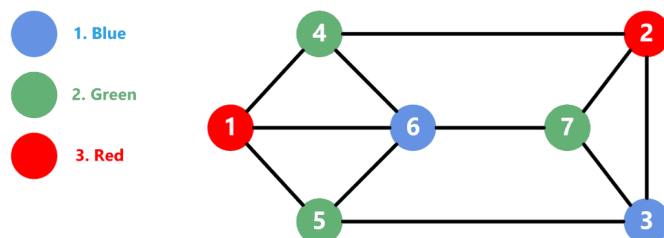
Implementation of a genetic algorithm (GA) to find the solution to the **Graph Coloring Problem (GCP)**.

The Graph Coloring Problem

Starting from a graph consisting of V vertices and E edges connecting them, we have to assign a color to each vertex of the graph such that the two following conditions are fulfilled:

- **No two adjacent vertices have the same color.** Remember that two vertices are considered adjacent if an edge connects them.
- **The number of colors used is the minimum possible.**

For example, let's consider the following graph, which complies with the two requirements described before:



You can get a more detailed explanation about the graph coloring problem from:
https://en.wikipedia.org/wiki/Graph_coloring

Genetic algorithm

Given a graph described by a list of nodes and links, the objective is the implementation of a genetic algorithm that obtains the value of the color of each node that matches with the two constraints introduced previously.

For the activity, you will have to take into account these points:

- How to codify the problem into a chromosome for the genetic algorithm
- How to validate that a chromosome represents a valid solution
- How to compute/measure the best solution.

In this activity you will also need to think about how to select the appropriate methods for crossover and mutation that will work with your description and codification of the problem.

Data and libraries

There are many examples of datasets that can be used. I recommend to use the ones of this link, since they also include in some cases the optimal solution:

<https://mat.tepper.cmu.edu/COLOR/instances.html>

You can use other datasets available from the internet that have the same structure for the problem.

Optional part: Implementation of two more optimization algorithms (3 points)

If you want to obtain the full grade, you should implement two more optimization methods to try to reach the best possible solution. In this case you can choose any of the methods that we have seen in class (tabu search, simulated annealing, extremal optimization, ...).

In this case you should be able to reuse most of the code done for the Genetic Algorithm (codification of the problem, total time computation, etc...), so you can focus on the implementation of the other method.

Delivery of the activity

This assignment can be done **alone or in pairs** (groups of two)

For this activity you must deliver **one PDF document** that includes:

- A link to the Github repository where the code of all the activity is accessible. More details on the code in the following sections. Remember that we require a history of all the commits, not only the last one with the final code.
- The name of the file should be **A2-Name1Surname1-Name2Surname2.pdf**

Also, in the document you must include the following two sections:

1. **Description of the chromosome, and adaptations done to the algorithm:**
Explain with detail how you translate the problem into the chromosome, and the different techniques that you have implemented for selection, mutation and crossover. You must AT LEAST implement two different techniques for each of those. You can also use elitism in the code if you think.

You must also explain how you choose the size of the population and how you identify that the system reaches a stationary state.

2. **The results of executing the code for 3 problems of different sizes:** We recommend running at least one problem with around 10 nodes, one with around 50-100 nodes and one with a larger graph (for instance, 400 or more nodes. Optionally you can try with even larger problems, since we will consider the complexity of the selected datasets (i.e. the number of links is also important!).

For each of the executions you should include in the document:

- a. The description of the dataset (URL where the data was collected, explanation of the data).
 - b. The results obtained with at least 6 different combinations of parameters described in the previous section.
 - c. For the best solution, present a figure with the evolution of the fitness in each of the algorithm steps.
3. **In case you implement more optimization methods**, you should describe the methods used, the parameters of the methods (what parameters have been used and how they have been chosen), and a comparison of the results obtained versus the genetic algorithm (in this case, it can be interesting to compare not only if the results are achieved, but how much time it takes to each algorithm / how many runs to get to the best result).